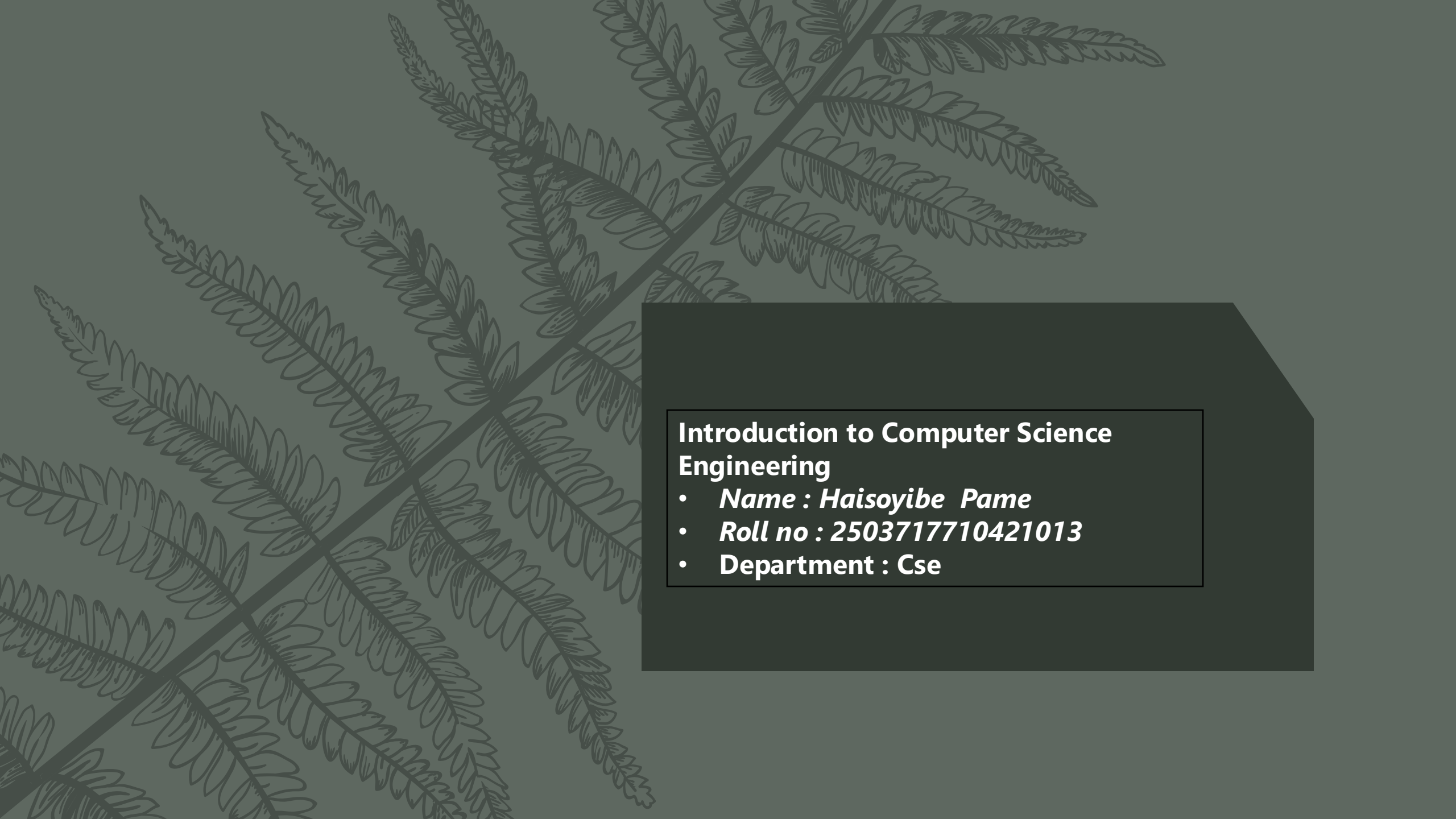




Introduction to Computer Science Engineering

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- ***Department : Cse***

1. What is Engineering?

Definition:

Engineering is the application of scientific and mathematical knowledge to design, build, and improve machines, systems, structures, and technologies that solve real-world problems.

Examples

- Building bridges
- Developing software
- Designing smartphones
- Constructing roads  

2. Difference Between Science and Engineering

Science	Engineering
Studies natural phenomena	Applies scientific knowledge
Focuses on discovering facts	Focuses on solving problems
Answers "Why?"	Answers "How?"
Research-oriented	Design-oriented
Example: Physics	Example: Civil Engineering

3. Difference Between Scientist and Engineer

Scientist	Engineer
Discovers new knowledge	Applies knowledge to solve problems
Conducts research and experiments	Designs and builds products or systems
Focuses on "Why?" Focuses on "How?"	
Develops theories	Creates practical solutions
Example: Albert Einstein	Example: Computer Software Engin

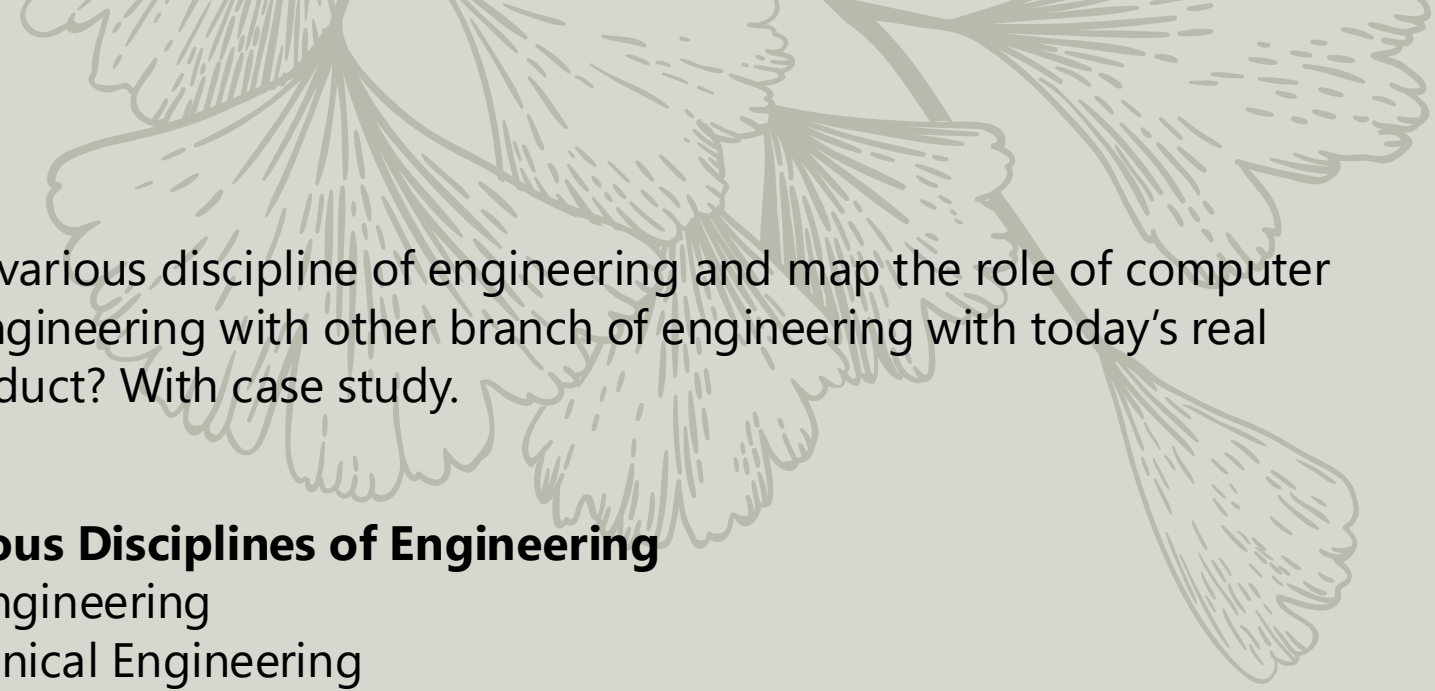
4. Need and Importance of Computer Science Engineering in Today's World

Ans: Computer Science Engineering is needed because it:

- Develops software and mobile applications.
- Supports Artificial Intelligence (AI).
- Provides Cybersecurity.
- Enables Cloud Computing.
- Builds Smart Cities and IoT devices.
- Improves healthcare through digital technologies.
- Supports online education and e-commerce.

Examples

- Google Search
- UPI payments
- Online banking
- Telemedicine
- Self-driving cars



5. List out various disciplines of engineering and map the role of computer science engineering with other branches of engineering with today's real world product? With case study.

Ans: Various Disciplines of Engineering

1. Civil Engineering
2. Mechanical Engineering
3. Electrical Engineering
4. Electronics and Communication Engineering (ECE)
5. Computer Science Engineering (CSE)
6. Chemical Engineering
7. Aerospace Engineering
8. Biomedical Engineering
9. Environmental Engineering
10. Artificial Intelligence & Data Science Engineering

Role of CSE with Other Engineering Branches

Branch	Role of Cse	Real-World Product
Civil	Smart building software	Smart Buildings
Mechanical	Robotics & Automation	Industrial Robots
Electrical	Smart energy systems	Electric Vehicles
Electronics	Embedded systems	Smartphones
Biomedical	Medical software	MRI Machines
Aerospace	Flight control software	Drones
Environmental	Pollution monitoring	Air Quality Systeme

Case Study – Tesla Electric Car

- **Product:** Tesla Electric Vehicle (EV)
Mechanical → Car design
- Electrical → Battery & Motor
- Electronics → Sensors & Cameras
- **Computer Science** → **AI, Autopilot & Software**
- Industrial → Manufacturing



6. Misconceptions About Computer Science Engineering

Ans:

- **✗ CSE is only about coding.**
✓ It also includes AI, Cybersecurity, Data Science, Networking, and Cloud Computing.
- **✗ CSE students only repair computers.**
✓ They develop software, apps, websites, and intelligent systems.
- **✗ Only math geniuses can study CSE.**
✓ Logical thinking, practice, and problem-solving are more important.
- **✗ CSE engineers work alone.**
✓ They work in teams with designers, engineers, and business professionals.
- **✗ CSE jobs are limited to IT companies.**
✓ CSE professionals work in healthcare, banking, education, automotive, gaming, and many other industries.



7. Skills Required for Computer Science Engineering in the 21st Century

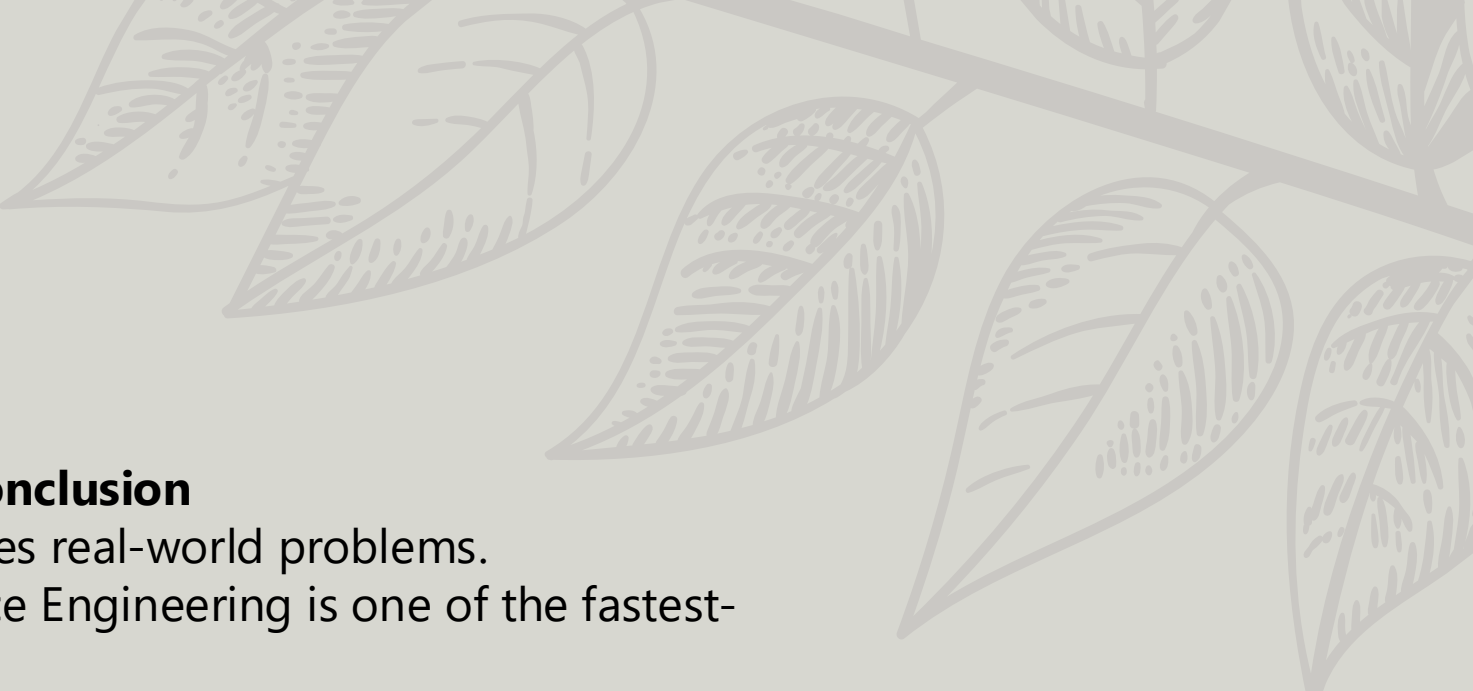
Ans:

Technical Skills

- Programming (Python, Java, C++)
- Data Structures & Algorithms
- Artificial Intelligence
- Cloud Computing
- Cybersecurity
- Database Management
- Web & Mobile Development

Soft Skills

- Problem-solving
- Communication
- Teamwork
- Leadership
- Creativity
- Time Management
- Critical Thinking
- Adaptability



Conclusion

- Engineering solves real-world problems.
- Computer Science Engineering is one of the fastest-growing fields.
- It works with almost every engineering discipline.
- Future technologies such as AI, Robotics, IoT, and Smart Cities depend on Computer Science Engineers.
- Continuous learning and innovation are essential for success.

The background of the slide features a stylized, hand-drawn illustration of several long, narrow leaves. These leaves are rendered in a light gray tone against a darker gray background. Each leaf is filled with fine, parallel lines, giving them a textured appearance. The leaves are arranged in a way that suggests they are part of a larger plant, with some overlapping others. The overall style is minimalist and artistic.

Thank You

Thank You!

Any Questions?